

UF2-NF1-Activitat 4. ACTIVITY LOGIN

Objectiu: Implementa la interfície gràfica que teniu a la pàgina pàgina 60 del llibre i connecta't al teu odoo i mostra que la connexió ha estat vàlida.

Per realitzar aquesta tasca primer mostraré el codi necessari.

- **Configuració de Retrofit i OkHttp:**

Utilitzo Retrofit per a les crides API i Gson per al maneig de JSON.

Incloc un interceptor de registre per facilitar la depuració.

- **OdooApiService i OdooClient:**

Defineix el punt final d'autenticació (/web/session/authenticate).

Configura el client amb temps d'espera adequats.

Proporciona un mètode per crear la sol·licitud d'autenticació en el format esperat per Odoo.

```
package com.example.myapplication;

import com.google.gson.JsonObject;
import retrofit2.Call;
import retrofit2.http.Body;
import retrofit2.http.POST;

public interface OdooApiService {
    @POST("/web/session/authenticate")
    Call<JsonObject> authenticate(@Body JsonObject loginRequest);
}
```

y_login.xml LoginActivity.java build.gradle.kts (:app) OdooApiService.java

```
1 package com.example.myapplication;
2
3 import com.google.gson.JsonObject;
4 import retrofit2.Call;
5 import retrofit2.http.Body;
6 import retrofit2.http.POST;
7
8 public interface OdooApiService {
9     @POST("/web/session/authenticate")
10     Call<JsonObject> authenticate(@Body JsonObject loginRequest);
11 }
12
```

```

package com.example.myapplication;

import com.google.gson.JsonObject;
import okhttp3.OkHttpClient;
import okhttp3.logging.HttpLoggingInterceptor;
import retrofit2.Retrofit;
import retrofit2.converter.gson.GsonConverterFactory;
import java.util.concurrent.TimeUnit;

public class OdooClient {
    private static final String BASE_URL = "http://192.168.1.129";
    private static OkHttpClient okHttpClient;
    private static Retrofit retrofit;
    private static OdooApiService apiService;

    private static OkHttpClient getOkHttpClient() {
        if (okHttpClient == null) {
            HttpLoggingInterceptor loggingInterceptor = new
HttpLoggingInterceptor();
loggingInterceptor.setLevel(HttpLoggingInterceptor.Level.BODY);

            okHttpClient = new OkHttpClient.Builder()
                .addInterceptor(loggingInterceptor)
                .connectTimeout(30, TimeUnit.SECONDS)
                .readTimeout(30, TimeUnit.SECONDS)
                .build();
        }
        return okHttpClient;
    }

    private static Retrofit getRetrofit() {
        if (retrofit == null) {
            retrofit = new Retrofit.Builder()
                .baseUrl(BASE_URL)
                .client(getOkHttpClient())
                .addConverterFactory(GsonConverterFactory.create())
                .build();
        }
        return retrofit;
    }

    public static OdooApiService getApiService() {
        if (apiService == null) {
            apiService =
getRetrofit().create(OdooApiService.class);
        }
        return apiService;
    }

    public static JsonObject createAuthRequest(String db, String

```

```

username, String password) {
    JSONObject params = new JSONObject();
    params.addProperty("db", db);
    params.addProperty("login", username);
    params.addProperty("password", password);

    JSONObject request = new JSONObject();
    request.add("params", params);

    return request;
}
}

```

Activity.java build.gradle.kts (:app) OdooApiService.java OdooClient.java ×

```

1 package com.example.myapplication;
2
3 import com.google.gson.JsonObject;
4 import okhttp3.OkHttpClient;
5 import okhttp3.logging.HttpLoggingInterceptor;
6 import retrofit2.Retrofit;
7 import retrofit2.converter.gson.GsonConverterFactory;
8 import java.util.concurrent.TimeUnit;
9
10 public class OdooClient {
11     private static final String BASE_URL = "http://192.168.1.129";
12     private static OkHttpClient okHttpClient;
13     private static Retrofit retrofit;
14     private static OdooApiService apiService;
15
16     private static OkHttpClient getOkHttpClient() {
17         if (okHttpClient == null) {
18             HttpLoggingInterceptor loggingInterceptor = new HttpLoggingInterceptor();
19             loggingInterceptor.setLevel(HttpLoggingInterceptor.Level.BODY);
20
21             okHttpClient = new OkHttpClient.Builder()
22                 .addInterceptor(loggingInterceptor)
23                 .connectTimeout(30, TimeUnit.SECONDS)
24                 .readTimeout(30, TimeUnit.SECONDS)
25                 .build();
26         }
27         return okHttpClient;
28     }

```

- **OdooSessionManager:**

Gestiona la sessió de l'usuari utilitzant SharedPreferences.

Emmagatzema l'ID de sessió, l'ID d'usuari i el nom d'usuari.

Proporciona mètodes per verificar si l'usuari està autenticat.

```
package com.example.myapplication;

import android.content.Context;
import android.content.SharedPreferences;

public class OdooSessionManager {
    private SharedPreferences sharedPreferences;
    private SharedPreferences.Editor editor;

    private static final String PREF_NAME = "OdooPrefs";
    private static final String KEY_SESSION_ID =
"session_id";
    private static final String KEY_USER_ID = "user_id";
    private static final String KEY_USERNAME = "username";
    private static final String KEY_IS_LOGGED_IN =
"is_logged_in";

    public OdooSessionManager(Context context) {
        sharedPreferences =
context.getSharedPreferences(PREF_NAME,
Context.MODE_PRIVATE);
        editor = sharedPreferences.edit();
    }

    public void saveSession(String sessionId, int userId,
String username) {
        editor.putString(KEY_SESSION_ID, sessionId);
        editor.putInt(KEY_USER_ID, userId);
        editor.putString(KEY_USERNAME, username);
        editor.putBoolean(KEY_IS_LOGGED_IN, true);
        editor.apply();
    }

    public void clearSession() {
        editor.clear();
        editor.apply();
    }

    public boolean isLoggedIn() {
        return sharedPreferences.getBoolean(KEY_IS_LOGGED_IN,
false);
    }

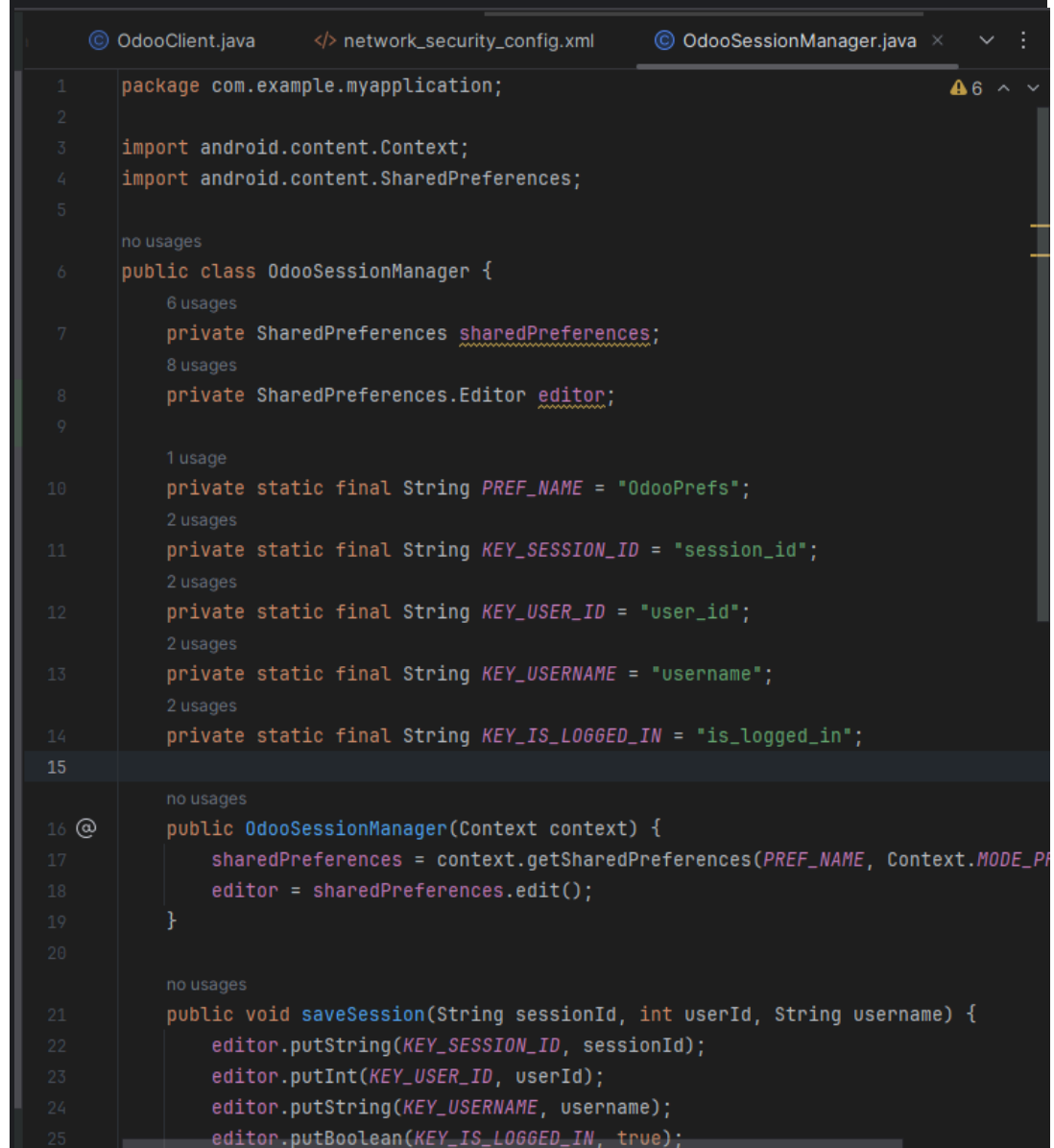
    public String getSessionId() {
        return sharedPreferences.getString(KEY_SESSION_ID,
null);
    }
}
```

```

    public int getUserId() {
        return sharedPreferences.getInt(KEY_USER_ID, -1);
    }

    public String getUsername() {
        return sharedPreferences.getString(KEY_USERNAME,
null);
    }
}

```



```

OdooClient.java  network_security_config.xml  OdooSessionManager.java
1  package com.example.myapplication;
2
3  import android.content.Context;
4  import android.content.SharedPreferences;
5
6  no usages
7  public class OdooSessionManager {
8      6 usages
9      private SharedPreferences sharedPreferences;
10     8 usages
11     private SharedPreferences.Editor editor;
12
13     1 usage
14     private static final String PREF_NAME = "OdooPrefs";
15     2 usages
16     private static final String KEY_SESSION_ID = "session_id";
17     2 usages
18     private static final String KEY_USER_ID = "user_id";
19     2 usages
20     private static final String KEY_USERNAME = "username";
21     2 usages
22     private static final String KEY_IS_LOGGED_IN = "is_logged_in";
23
24     no usages
25     @ public OdooSessionManager(Context context) {
26         sharedPreferences = context.getSharedPreferences(PREF_NAME, Context.MODE_PRIVATE);
27         editor = sharedPreferences.edit();
28     }
29
30     no usages
31     public void saveSession(String sessionId, int userId, String username) {
32         editor.putString(KEY_SESSION_ID, sessionId);
33         editor.putInt(KEY_USER_ID, userId);
34         editor.putString(KEY_USERNAME, username);
35         editor.putBoolean(KEY_IS_LOGGED_IN, true);
36     }
37 }

```

- **UI d'inici de sessió:**

Una interfície senzilla amb camps per a la URL del servidor, base de dades, usuari i

contrasenya.

Inclou gestió d'errors i estats de càrrega.

```
package com.example.myapplication;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.ProgressBar;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import com.example.myapplication.R;
import com.example.myapplication.OdooClient;
import com.example.myapplication.OdooSessionManager;
import com.google.gson.JsonObject;
import com.google.gson.JsonObject;

import retrofit2.Call;
import retrofit2.Callback;
import retrofit2.Response;

public class LoginActivity extends AppCompatActivity {
    private EditText etServerUrl;
    private EditText etDatabase;
    private EditText etUsername;
    private EditText etPassword;
    private Button btnLogin;
    private ProgressBar progressBar;
    private OdooSessionManager sessionManager;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_login);

        // Inicializar vistas
        etServerUrl = findViewById(R.id.etServerUrl);
        etDatabase = findViewById(R.id.etDatabase);
        etUsername = findViewById(R.id.etUsername);
        etPassword = findViewById(R.id.etPassword);
        btnLogin = findViewById(R.id.btnLogin);
        progressBar = findViewById(R.id.progressBar);

        // Inicializar session manager
        sessionManager = new OdooSessionManager(this);

        // Verificar si ya hay sesión activa
        if (sessionManager.isLoggedIn()) {
            // Navegar a la actividad principal
            navigateToMainActivity();
        }
    }
}
```

```

        btnLogin.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                login();
            }
        });
    }

    private void login() {
        String serverUrl =
etServerUrl.getText().toString().trim();
        String database =
etDatabase.getText().toString().trim();
        String username =
etUsername.getText().toString().trim();
        String password = etPassword.getText().toString();

        if (serverUrl.isEmpty() || database.isEmpty() ||
username.isEmpty() || password.isEmpty()) {
            Toast.makeText(this, "Por favor, complete todos
los campos", Toast.LENGTH_SHORT).show();
            return;
        }

        // Mostrar progreso
        progressBar.setVisibility(View.VISIBLE);
        btnLogin.setEnabled(false);

        // Crear solicitud de autenticación
        JSONObject authRequest =
OdooClient.createAuthRequest(database, username, password);

        // Realizar la llamada de autenticación
OdooClient.getApiService().authenticate(authRequest).enqueue
(new Callback<JSONObject>() {
            @Override
            public void onResponse(Call<JSONObject> call,
Response<JSONObject> response) {
                progressBar.setVisibility(View.GONE);
                btnLogin.setEnabled(true);

                if (response.isSuccessful()) {
                    JSONObject responseJson =
response.body();
                    JSONObject result = null;

                    if (responseJson != null &&
responseJson.has("result")) {
                        result =
responseJson.getAsJsonObject("result");
                    }
                }
            }
        });
    }

```

```

        if (result != null &&
!result.has("error")) {
            // Extraer información de sesión
            int userId = -1;
            if (result.has("uid") &&
!result.get("uid").isJsonNull()) {
                userId =
result.get("uid").getAsInt();
            }

            String sessionId = "";
            if
(response.headers().get("set-cookie") != null) {
                sessionId =
response.headers().get("set-cookie");
            }

            if (userId > 0) {
                // Guardar información de sesión
sessionManager.saveSession(sessionId, userId, username);

                // Navegar a la actividad
principal
                navigateToMainActivity();
            } else {

Toast.makeText(LoginActivity.this,
                    "Credenciales
inválidas", Toast.LENGTH_SHORT).show();
            }
        } else {
            String errorMsg = "Error de
autenticación";

            if (result != null &&
result.has("error")) {
                JsonObject error =
result.getAsJsonObject("error");
                if (error.has("message")) {
                    errorMsg =
error.get("message").getString();
                }
            }
            Toast.makeText(LoginActivity.this,
errorMsg, Toast.LENGTH_SHORT).show();
        }
    } else {
        Toast.makeText(LoginActivity.this,
            "Error: " + response.code(),
Toast.LENGTH_SHORT).show();
    }
}

@Override
public void onFailure(Call<JsonObject> call,

```



```

Throwable t) {
    progressBar.setVisibility(View.GONE);
    btnLogin.setEnabled(true);
    Toast.makeText(LoginActivity.this,
        "Error de conexión: " +
t.getMessage(), Toast.LENGTH_SHORT).show();
    }
    });
}

private void navigateToMainActivity() {
    Toast.makeText(this, "Inicio de sesión exitoso",
Toast.LENGTH_SHORT).show();
    }
}

```

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    android:gravity="center">

    <EditText
        android:id="@+id/etServerUrl"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="URL del servidor"
        android:inputType="textUri"
        android:layout_marginBottom="8dp"/>

    <EditText
        android:id="@+id/etDatabase"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Base de datos"
        android:layout_marginBottom="8dp"/>

    <EditText
        android:id="@+id/etUsername"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Nombre de usuario"
        android:inputType="textEmailAddress"
        android:layout_marginBottom="8dp"/>

    <EditText
        android:id="@+id/etPassword"

```

```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Contraseña"
        android:inputType="textPassword"
        android:layout_marginBottom="16dp"/>
```

```
<Button
```

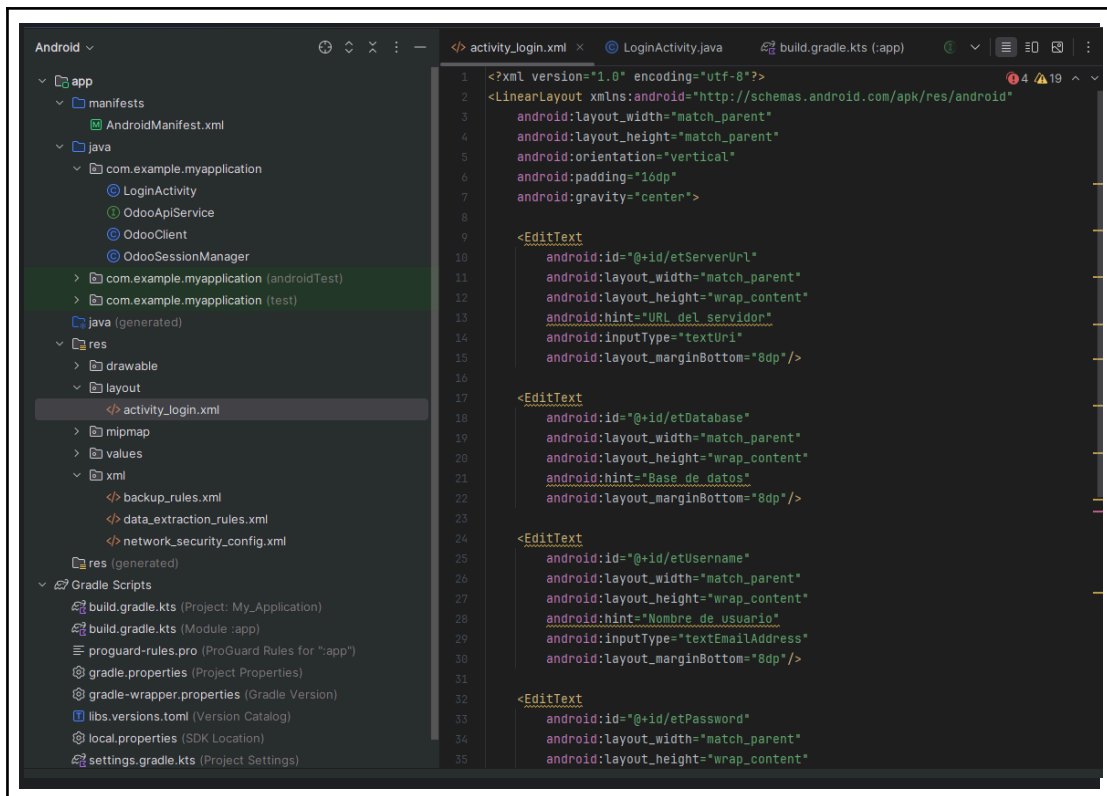
```
    android:id="@+id/btnLogin"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Iniciar sesión"/>
```

```
<ProgressBar
```

```
    android:id="@+id/progressBar"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="16dp"
    android:visibility="gone"/>
```

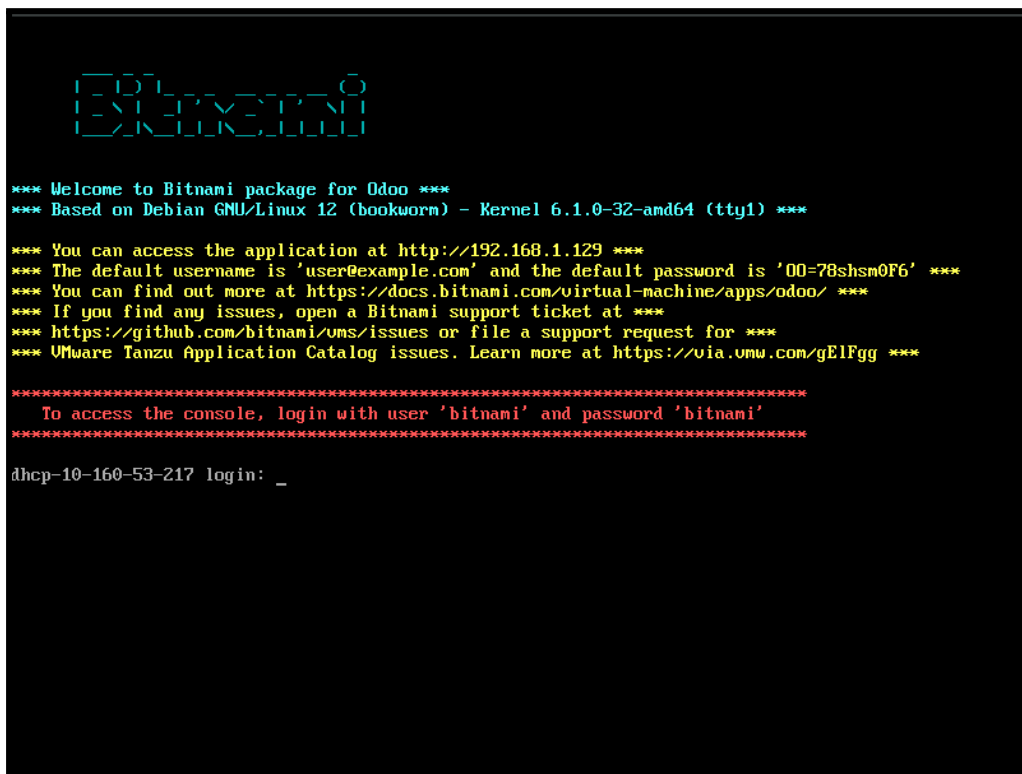
```
</LinearLayout>
```

```
22     public class LoginActivity extends AppCompatActivity {
32         protected void onCreate(Bundle savedInstanceState) {
53             btnLogin.setOnClickListener(new View.OnClickListener() {
57                 }
58             });
59         }
60
61         1 usage
62         private void login() {
63             String serverUrl = etServerUrl.getText().toString().trim();
64             String database = etDatabase.getText().toString().trim();
65             String username = etUsername.getText().toString().trim();
66             String password = etPassword.getText().toString();
67
68             if (serverUrl.isEmpty() || database.isEmpty() || username.isEmpty() ||
69                 password.isEmpty()) {
70                 Toast.makeText(context, this, text: "Por favor, complete todos los campos",
71                     duration: Toast.LENGTH_SHORT).show();
72                 return;
73             }
74
75             // Mostrar progreso
76             progressBar.setVisibility(View.VISIBLE);
77             btnLogin.setEnabled(false);
78
79             // Crear solicitud de autenticación
80             JSONObject authRequest = Odooclient.createAuthRequest(database, username, password);
81
82             // Realizar la llamada de autenticación
83             Odooclient.getApiService().authenticate(authRequest).enqueue(new CallBack() {
84                 @Override
85                 public void onResponse(Call<JsonObject> call, Response<JsonObject> response) {
86                     progressBar.setVisibility(View.GONE);
87                     btnLogin.setEnabled(true);
88
89                     if (response.isSuccessful()) {
90                         JsonObject responseJson = response.body();
91                     }
92                 }
93             });
94         }
95     }
96 }
```

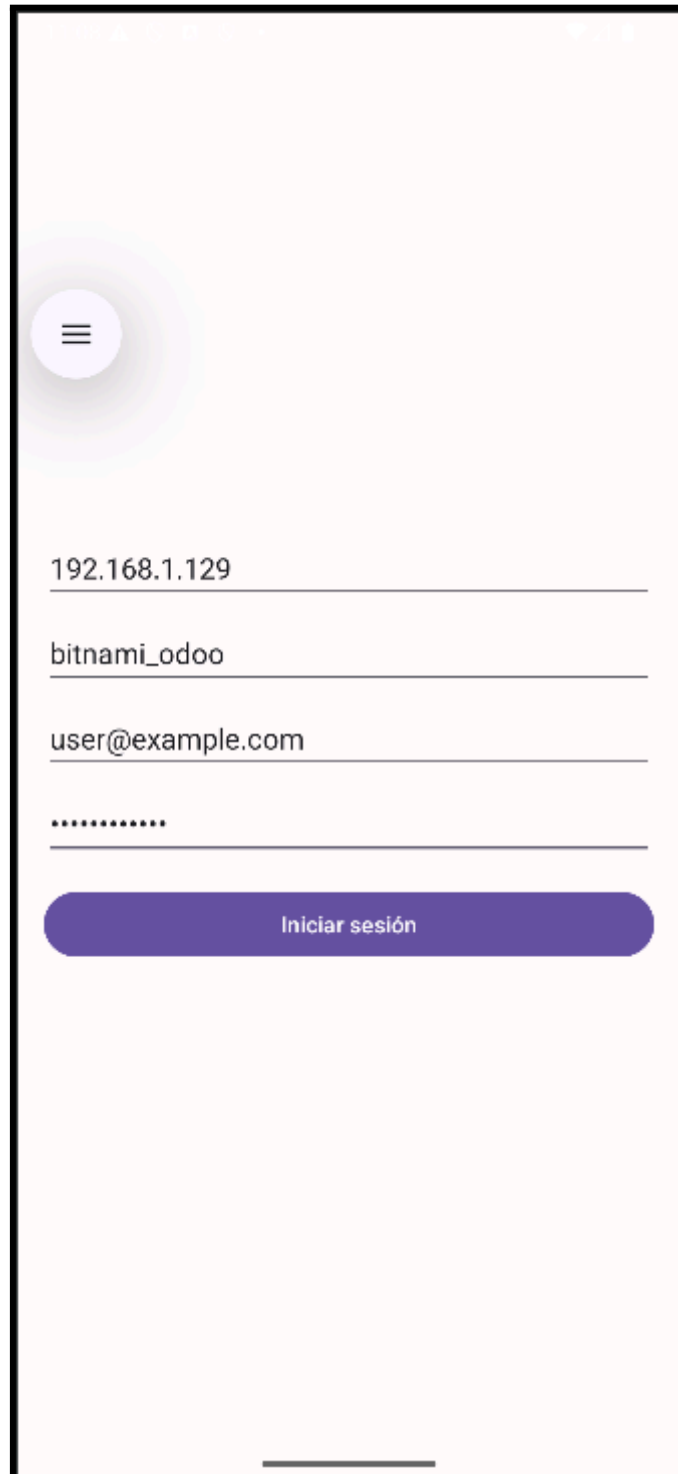


Proves

Màquina bitnami



Login



A mobile application login screen with a light pink background. At the top left, there is a circular menu icon with three horizontal lines. Below it, there are four input fields: the first contains the IP address '192.168.1.129', the second contains 'bitnami_odoo', the third contains 'user@example.com', and the fourth contains a series of dots representing a password. Below the password field is a purple rounded rectangular button with the text 'Iniciar sesión'. At the bottom center, there is a horizontal line representing a mobile home indicator.

192.168.1.129

bitnami_odoo

user@example.com

.....

Iniciar sesión

192.168.1.129

bitnami_odoo

user@example.com

Iniciar sesión

Inicio de sesión exitoso

```
2025-04-05 13:08:26.086 16395-16758 okhttp.OkHttpClient com.example.myapplication
2025-04-05 13:08:26.086 16395-16758 okhttp.OkHttpClient com.example.myapplication
2025-04-05 13:08:26.086 16395-16758 okhttp.OkHttpClient com.example.myapplication
2025-04-05 13:08:26.087 16395-16758 okhttp.OkHttpClient com.example.myapplication
2025-04-05 13:08:26.087 16395-16758 okhttp.OkHttpClient com.example.myapplication
2025-04-05 13:08:26.840 16395-16758 okhttp.OkHttpClient com.example.myapplication
2025-04-05 13:08:26.842 16395-16758 okhttp.OkHttpClient com.example.myapplication
2025-04-05 13:08:26.843 16395-16758 okhttp.OkHttpClient com.example.myapplication
2025-04-05 13:08:26.846 16395-16758 okhttp.OkHttpClient com.example.myapplication
2025-04-05 13:08:26.846 16395-16758 okhttp.OkHttpClient com.example.myapplication
2025-04-05 13:08:26.846 16395-16758 okhttp.OkHttpClient com.example.myapplication
2025-04-05 13:08:26.846 16395-16758 okhttp.OkHttpClient com.example.myapplication
2025-04-05 13:08:26.846 16395-16758 okhttp.OkHttpClient com.example.myapplication
2025-04-05 13:08:26.846 16395-16758 okhttp.OkHttpClient com.example.myapplication
2025-04-05 13:08:26.846 16395-16758 okhttp.OkHttpClient com.example.myapplication
2025-04-05 13:08:26.847 16395-16758 okhttp.OkHttpClient com.example.myapplication
2025-04-05 13:08:26.847 16395-16758 okhttp.OkHttpClient com.example.myapplication
```

```
--> POST http://192.168.1.129/web/session/authenticate
Content-Type: application/json; charset=UTF-8
Content-Length: 85
{"params":{"db":"bitnami_odoo","login":"user@example.com","password":"00=78shsm0F6"}}
--> END POST (85-byte body)
<-- 200 OK http://192.168.1.129/web/session/authenticate (753ms)
Date: Sat, 05 Apr 2025 11:08:44 GMT
Server: Werkzeug/3.0.1 Python/3.12.9
Content-Length: 2007
Content-Type: application/json; charset=utf-8
X-Content-Type-Options: nosniff
Set-Cookie: session_id=rjbpd_PJq0Y-8v6udZR6dV4ja4BHS7hrJBz3gbjb5t6WRK6NCwgca6CV-pNvSW_UAmw6owAdJ0a_63R40HfM; Expires=Sun,
Keep-Alive: timeout=2, max=100
Connection: Keep-Alive
{"jsonrpc": "2.0", "id": null, "result": {"uid": 2, "is_system": true, "is_admin": true, "is_public": false, "is_internal_
<-- END HTTP (2007-byte body)
```